Module -3

① Explain Matplotlib API, with suitable examples?

Matplotlib API :-

Matplotlib is a popular python library used creating graphs, charts & data visualization. It is widely used in data science.

The matplotlib API provides functions & classes :-

Line plots
Bar charts
Histograms
Scatter plots
Pie charts
Bar plots

function	purpose
plt.plot()	create a line plot
plt.bar()	create a bar chart
plt.show()	create a scatter plot
plt.hist()	create a histogram
plt.pie()	create a piechart

Basic structure of matplotlib :-

1. Import matplotlib
2. prepare the data
3. select the type of plot
4. Add filters and labels
5. customize the plot if required

Advantages :-

1. Easy to use
2. Many plot types
3. Highly customization
4. Works with numpy & pandas.

(2) Different line plot, Barplot, Histograms & Scatter plot.

Different types of plot are used for different purpose in data visualization.

Line plots
Bar plots
Histograms
Scatter plots
four commonly used plots

1. Line plot :-

A plot represents data points connected by lines mainly used to show trends or changes an ordered sequence.

```
import matplotlib.pyplot as plt
months = ["Jan", "Feb", "mar", "Apr"]
plt.plot(months, sales)
plt.show()
```

2. Bar plot :-

A Bar plot represents data using rectangular bars. The height or length of each bar represents the value of a category.

3. histogram :-
A histogram is used to represent the distribution of numerical data. It divides data into intervals called bins & shows how many observations fall into each interval.

4. scatter plot :-
A scatter plot displays individual data points using dots. It is mainly used to examine the relationship or correlation b/n numerical variables.

(3) Explain seaborn facet grid & categorical plot

seaborn :-
seaborn is a python data visualization lib built on top of matplotlib. It provides a high-level interface for creating attractive & informative.

→ It is commonly imported as
import seaborn as sns
Two imp concepts in seaborn are
1. Face grid
2. categorical plots

seaborn facegrid :-
facegrid is a seaborn features used to create multiple plots based on different subsets of a dataset.

Syntax :-
sns.facegrid(data, col = "column")
Here data = dataset
col = variable used to create columns.

Ex:-
suppose we have a dataset containing
- Student
- Gender
- Marks
- Department

Important facegrid parameter

parameter	purpose
Data	Dataset
Row	creates, separates rows based on a variable
column	creates separate columns
Hue	uses different colours for categories.